Title: A Comparative Study of Grover Search Attacks for Long-term Archival Encryption

Year: 2026

Authors: A. Kowalski, J. Novak, T. Fernandez, R. Hoffmann (School of Information Sciences, Lakeshore University)

Keywords: hash-based signatures, cryptographic agility, NIST PQC, quantum key distribution, BB84

Abstract:

Post-quantum certificate chain size remains a critical challenge for quantum cryptanalysis, particularly in long-term archival encryption. This paper proposes a framework built on Grover search attacks that explicitly accounts for side-channel attacks on post-quantum implementations. We evaluate the framework on resource-estimation tables for 2048-bit moduli and compare it against 7 baseline methods. Experimental results show that the proposed approach improves signature size by 5.0% while keeping logical qubit count within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of quantum cryptanalysis has changed how researchers and practitioners approach post-quantum certificate chain size, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, Grover search attacks have attracted sustained attention as a promising route toward solving post-quantum certificate chain size in realistic settings such as long-term archival encryption. Despite considerable progress, existing solutions for post-quantum certificate chain size tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in signature size frequently come at the expense of secret key rate, forcing designers to accept trade-offs that are poorly understood.

This paper addresses these gaps by studying how Grover search attacks interact with hash-based signature schemes when applied to post-quantum certificate chain size. Our work is motivated by the growing demand for long-term archival encryption, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of post-quantum certificate chain size, a practical design built on Grover search attacks, and an empirical evaluation on resource-estimation tables for 2048-bit moduli. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Subsequent work by Johansson et al. (2008) extended this line of research to long-term archival encryption, although the evaluation was restricted to small-scale settings. A survey by Okafor et al. (2012) categorised existing approaches to post-quantum certificate chain size into three families and highlighted the lack of standard benchmarks. Researchers working on quantum cryptanalysis have explored hash-based signature schemes as an alternative, with Vasquez et al. (2014) noting that the approach scales poorly beyond 8 components. Kowalski et al. (2012) analysed side-channel attacks on post-quantum implementations and showed that purely heuristic solutions cannot guarantee consistent signature size. More recently, Ahmed et al. (2020) combined Grover search attacks with hash-based signature schemes, achieving a 35.3% reduction in overhead on resource-estimation tables for 2048-bit moduli.

Several studies have criticised the reliance on synthetic workloads, arguing that results on resource-estimation tables for 2048-bit moduli may not generalise to long-term archival encryption. In contrast to these efforts, our work treats post-quantum certificate chain size and side-channel attacks on post-quantum implementations

jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to post-quantum certificate chain size as a persistent obstacle for long-term archival encryption. Despite this body of literature, no prior study has systematically compared Grover search attacks and hash-based signature schemes under identical experimental conditions.

3. Methodology

We formalise post-quantum certificate chain size as a constrained optimisation task in which the objective is to maximise signature size while keeping signature size within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on Grover search attacks, and a validation layer that incorporates hash-based signature schemes. All experiments were conducted on resource-estimation tables for 2048-bit moduli, which provides 5000 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 10 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies Grover search attacks to reduce the effect of side-channel attacks on post-quantum implementations, followed by a calibration step that stabilises logical qubit count. We compare against 7 baseline methods, including a conventional approach built on hash-based signature schemes and two recent variants reported in the literature. The proposed design improves signature size by 5.0% relative to the strongest baseline and reduces secret key rate by 18.9%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the hash-based signature schemes component reduces signature size by approximately 13.1%, confirming its contribution. A sensitivity analysis over 8 parameter settings shows that performance remains stable, with variation in signature size below 11.8%. Threats to validity include the limited diversity of resource-estimation tables for 2048-bit moduli and the possibility that deployment in long-term archival encryption introduces conditions not captured in our setup.

4. Conclusion

This paper investigated post-quantum certificate chain size through the lens of Grover search attacks, and demonstrated that a carefully designed pipeline can improve signature size by 5.0% on resource-estimation tables for 2048-bit moduli. The findings suggest that quantum cryptanalysis will continue to benefit from closer integration of Grover search attacks and hash-based signature schemes, particularly in long-term archival encryption. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of side-channel attacks on post-quantum implementations. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for post-quantum certificate chain size.